Build a Memory-First Web Agent in Python

Design and implement a GenAI agent that:

1. Answers user questions by **checking a Redis memory first** (vector search over embedded data).
2. **Falls back to the web** when memory misses, then ingests what it finds into Redis for future reuse.
3. Returns a grounded answer with clear metadata (URLs) to the sources used.

- You may implement using **LangGraph**, **LangChain**, **MS Agent Freamwork**, or **plain Python**. Python is mandatory; framework choice is up to you.
- You could use AI assistance GitHub CoPilot, if you did, please include all instructions and explanation on how you have used code assistance.
- You are **not required** to host or deploy the solution, please share the repo only. Functional requirements (what it must do)

Memory-first routing:

- Embed the user query, perform **vector search in Redis**.
- If top result similarity ≥ threshold (default 0.7), **answer from memory only** (include stored metadata).
- Else **search the web**, fetch top pages, summarize, **store chunks & metadata in Redis**, then answer using retrieved context.

Web search & fetch:

- Use any web search APIs
- Fetch web pages content after search.
- Convert webpages into a markdown content.

LLMs:

- Use 2 llm of your choice(one for conversation and one for analytics), however provide explanation of your choice, cost & quality.

Log:

- Log each turn was it a memory hit or memory miss + WebSearch

Analytics:

- Provide analytics on what type of topics and questions users asked.

Non-functional requirements

- **Security**: Basic prompt-injection guardrails
- **Reliability**: Timeouts/retries for network or token issues.